Evaluating Feature-Based Models for Post-Disaster Satellite Image Classification

Shou-Jen Chen [*1] **Inayyah Don Nazwim** [*1]

Abstract

In this work, we evaluate interpretable, feature-based machine learning pipelines for post-disaster satellite image classification on the xView2 dataset, addressing two tasks: binary disaster-type classification (Midwest Flooding vs. SoCal Fire) and four-class damage severity classification within the Hurricane Matthew subset. Rather than relying on deep neural networks, we extract handcrafted features from RGB color statistics, Sobel edge detection, Local Binary Patterns (LBP), and Gabor filter responses, and use Cohen's d to systematically measure their discriminative power before model training. LBP texture features dominate both tasks; logistic regression achieves 99.0% cross-validated accuracy on Task A, while the best damage severity model (soft-voting ensemble of Logistic Regression, Random Forest, and XGBoost) reaches a macro-F1 of 0.501 ± 0.040, limited primarily by extreme class imbalance (label 2 has only 7 training examples in the Hurricane Matthew subset). These results show that texture features enable near-perfect disaster-type separation, while fine-grained damage severity classification remains an open challenge requiring richer representations or additional data. Youtube link: `https://youtu.be/tSi-gjOoNSM`

1. Introduction

Rapid and accurate assessment of damage caused by natural disasters is critical for coordinating emergency response. Satellite imagery provides one of the few scalable means of surveying large affected areas quickly, yet manual annotation of thousands of image tiles is both time-consuming and expensive. Automated classification systems that can ingest post-disaster satellite images and output disaster type or damage severity labels would allow response agencies to allocate resources far more efficiently.

Existing work has largely focused on deep learning. The xBD benchmark (Gupta et al., 2019) introduced a large-scale annotated dataset and showed that CNN-based architectures, including U-Net variants, achieve strong performance on building-level damage segmentation. Subsequent work, including top xView2 competition entries, relied on multi-task learning, pre-trained encoders, and ensemble strategies (Gupta et al., 2019). While impressive, these approaches require large annotated datasets, significant compute, and offer limited interpretability.

A complementary line of research asks whether simpler, interpretable features can achieve competitive performance. Classical computer vision descriptors such as Local Binary Patterns (Ojala et al., 2002) and Gabor filters (Manjunath & Ma, 1996) have long been used for texture classification and scene recognition. Our project sits within this tradition: we ask how far handcrafted, interpretable features can take us on the xView2 satellite image classification problem.

Our approach differs from existing work in two main respects. First, we avoid deep feature representations entirely, focusing on features whose connection to physical scene properties (texture roughness, edge density, and color hue) is transparent. Second, before training any classifier, we use Cohen's d as a systematic discriminability measure to compare feature categories and identify which physical aspects of images carry the most signal for each task. This feature-first methodology is central to our research questions:

1. Can handcrafted image features distinguish disaster types (Midwest Flooding vs. SoCal Fire)?
2. Can the same features distinguish damage severity levels (minor vs. destroyed within Hurricane Matthew)?

2. Description of Data

2.1. Dataset Overview

Our dataset is derived from the xView2/xBD challenge (Gupta et al., 2019). The full xBD training set contains 18,336 annotated satellite images spanning six disaster types; we use a curated sample of 200 images per disaster across three events (Hurricane Matthew, SoCal Fire, and Midwest Flooding), yielding 600 images total.

[*]Equal contribution [1]Department of EECS, University of California, Berkeley, U.S.. Correspondence to: Shou-Jen Chen <lawrancechen@berkeley.edu>, Inayyah Don Nazwim <inayyah_d@berkeley.edu>.

Each image is a post-disaster satellite tile stored as a NumPy array of shape $1024 \times 1024 \times 3$, with pixel values in $[0, 255]$. The granularity is one tile per observation, covering a fixed geographic area and assigned a single tile-level integer label representing the majority damage severity category (0: no damage, 1: minor damage, 2: major damage, 3: destroyed). Metadata columns include `n_buildings` (total buildings in the tile) and `majority_share` (proportion of buildings in the majority damage class).

2.2. Data Preparation and Cleaning

A validity check confirmed all 600 labels are integers in $\{0, 1, 2, 3\}$ with zero missing values. The `majority_share` column contains missing entries when `n_buildings` $= 0$; since our analysis uses only image-level features, this column was dropped. In total, 152 of 600 images have zero buildings, with SoCal Fire the highest at 55.5% of its tiles (111 of 200). These tiles contain only burnt terrain, sky, and vegetation rather than building structures, so per-pixel texture and edge statistics reflect ground cover rather than structural damage, confounding any feature that implicitly assumes buildings are present. They are retained because the absence of structures is itself a meaningful signal for disaster-type classification. No transformations were applied to the raw image arrays prior to feature extraction. Some xBD tiles contain zero-valued border padding; features were computed over the full 1024×1024 array, so per-pixel statistics such as color means, entropy, and edge density may be slightly biased downward in affected tiles. Features were standardized (zero mean, unit variance) at training time using `StandardScaler` fitted on the training fold only, preventing data leakage.

2.3. Class Distribution and Imbalance

At the disaster-type level the three disasters are balanced (200 images each). At the damage label level, severe imbalance exists (Table 1). Label 2 has only 12 samples total (6 in Hurricane Matthew, 6 in SoCal Fire) and zero in Midwest Flooding. Labels 0 and 3 together account for $\approx 85\%$ of observations, meaning a majority-class classifier would score 46.5% accuracy while being entirely useless for minority classes.

Disaster	0	1	2	3
Hurricane Matthew	31	70	6	93
SoCal Fire	137	7	6	50
Midwest Flooding	26	84	0	90
Total	194	161	12	233

Table 1. Damage label counts per disaster. Label 2 is critically scarce.

2.4. Feature Engineering

Each $1024 \times 1024 \times 3$ image was reduced to a row of scalar features through four extraction pipelines, converting unstructured arrays into structured input for classical classifiers:

Color (RGB) features. Per-channel mean and standard deviation (`mean_r`, `mean_g`, `mean_b`, `std_r`, `std_g`, `std_b`). Brightness is the mean intensity across all channels; contrast is the standard deviation across all channels.

Sobel edge features. The Sobel operator computes the image gradient magnitude at each pixel, highlighting regions of sharp intensity change. `edge_mean` is the overall edge response; `edge_std` is variability in edge strength; `edge_density` is the proportion of pixels exceeding a fixed magnitude threshold.

LBP texture features (Ojala et al., 2002). Local Binary Patterns encode local neighborhood structure by comparing each pixel to its circular neighborhood of radius r. High standard deviation of LBP codes indicates rough, complex texture (water surfaces, rubble); low values indicate smooth, uniform surfaces (charred terrain, intact rooftops). We extract the mean and standard deviation at radii $r \in \{1, 2, 3\}$, capturing texture at fine, intermediate, and coarse spatial scales (6 features total).

Gabor filter features (Manjunath & Ma, 1996). Gabor filters are oriented sinusoidal bandpass filters; a high response means the image contains spatial frequency content at the filter's orientation and scale. Intact buildings produce regular, grid-like edges at specific orientations; rubble produces isotropic, disorganized edges with no dominant orientation. We extract the mean and standard deviation of each kernel response across all combinations of orientations $\theta \in \{0°, 45°, 90°, 135°\}$ and spatial frequencies $f \in \{0.1, 0.2, 0.3\}$ (12 kernels, 24 features).

Semantic and texture features. Domain knowledge motivates eight additional features not captured by generic image statistics. `laplacian_var` (variance of the Laplacian) measures structural chaos: rubble and collapsed buildings produce high values due to many irregular high-frequency edges, and achieves the highest ANOVA-F for Task B ($F = 26.82$). `veg_index_mean` and `veg_index_std` approximate vegetation coverage from the green-to-red ratio. `blue_tarp_ratio` estimates the fraction of pixels consistent with FEMA blue tarps placed on damaged roofs after storms ($F = 5.83$ for Task B). `gray_pixel_ratio` captures low-saturation pixels (concrete, ash, metal). `image_entropy` measures overall information content of the grayscale histogram. `glcm_contrast`, `glcm_homogeneity`, `glcm_energy`, and `glcm_correlation` are Gray-Level Co-occurrence Matrix descriptors (Haralick et al.,

1973) that quantify spatial regularity: intact buildings are texturally regular, while rubble is disorganized.

The final feature vector is 58-dimensional: 6 color (RGB mean/std) + 3 Sobel + 6 LBP + 24 Gabor + 8 semantic/GLCM + 11 extended color statistics (per-channel IQR, HSV channel statistics).

3. Methodology

3.1. Overview

Our pipeline has four stages: (1) feature extraction from each raw image, (2) feature normalization, (3) class imbalance handling within CV folds, and (4) training and evaluation of classifiers. All stages are implemented in Python using scikit-image, scikit-learn, and imbalanced-learn.

3.2. Feature Extraction

We extract a 58-dimensional feature vector from each image (Section 2.4), computed on the full 1024×1024 tile and pre-computed once, then cached as CSV files to avoid re-running expensive computations. A 12-feature baseline subset is selected via Cohen's d analysis (described below) for interpretability comparisons; the full 58-feature set is used for all final classifiers.

3.3. Feature Selection via Cohen's d

We used Cohen's d as the primary feature selection criterion (Equation 1). For two classes with means μ_1, μ_2 and standard deviations σ_1, σ_2:

$$d = \frac{|\mu_1 - \mu_2|}{\sqrt{(\sigma_1^2 + \sigma_2^2)/2}} \tag{1}$$

Values $d > 0.8$ are conventionally a large effect size. We also computed the Pearson correlation matrix across all features; when two features within the same family had $|r| > 0.80$, we retained only the higher-d member.

The 12 features selected for the baseline model are: `mean_r`, `mean_g`, `mean_b`, `std_r`, `std_g`, `std_b` (color); `edge_mean`, `edge_std` (Sobel); `lbp_r1_std`, `lbp_r3_mean` (LBP, chosen for highest d per spatial scale while avoiding highly correlated pairs); and `gabor_t135_f01_mean`, `gabor_t135_f01_std` (highest-d Gabor configuration in both tasks). After observing suboptimal Task B performance with this baseline, we cross-validated the selection against Random Forest feature importances (Appendix C) and expanded to the full 58-dimensional feature set described in Section 2.4.

3.4. Task A: Disaster-Type Classification

Task A is a balanced binary problem (200 Midwest Flooding vs. 200 SoCal Fire). We use **Logistic Regression** because (1) the classes are nearly linearly separable in LBP space (`lbp_r1_std` achieves $d = 2.60$), and (2) the learned weight vector directly quantifies each feature's contribution, supporting interpretability. Performance is measured by **accuracy**, appropriate here due to class balance.

3.5. Task B: Damage Severity Classification

Task B is a four-class problem on the 200 Hurricane Matthew training images. With label 2 having only 6 training examples (3.0% of the training set), this is substantially harder. We evaluate three models:

Logistic Regression (`class_weight='balanced'`, `max_iter=1000`): a linear baseline with minority-class reweighting. Regularization strength tuned via iterative grid search: an initial coarse sweep over $C \in \{0.01, 0.1, 1, 10, 100\}$ identified the $[0.1, 1]$ region, which was progressively refined in subsequent runs; final best: $C = 0.6$ (CV macro-F1: 0.449).

Random Forest (`class_weight='balanced'`): an ensemble of decision trees that captures non-linear feature interactions. Hyperparameters were tuned via `GridSearchCV` over $\{$`n_estimators`$\} \in \{100, 200, 300\}$ and $\{$`max_depth`$\} \in \{10, 20, \text{None}\}$; best params: `n_estimators=100`, `max_depth=10`.

XGBoost (`eval_metric='mlogloss'`): gradient-boosted trees that iteratively correct residual errors. Tuned via `GridSearchCV` over `n_estimators` $\in \{200, 300\}$, `max_depth` $\in \{3, 4, 6\}$, `learning_rate` $\in \{0.05, 0.1\}$; best params: `n_estimators=300`, `max_depth=3`, `learning_rate=0.1`.

We also test a **soft-voting ensemble** of all three, averaging class probabilities before taking the argmax.

To address imbalance we apply **SMOTE** (Chawla et al., 2002) ($k_{\text{neighbors}} = 3$) within each cross-validation fold: synthetic minority samples are generated by interpolating between real examples and their k nearest neighbors. The default $k = 5$ would fail because label 2 has at most 4 examples per training fold (6 total $\times$ 80% train); we reduce to $k = 3$ to remain feasible. Concretely, with 6 label-2 examples under 5-fold stratified CV, each validation fold contains exactly 1 label-2 image and each training fold contains the remaining 5, making CV macro-F1 estimates for label 2 highly variable across folds. SMOTE is applied only to the training split (after scaling), never to validation, preventing leakage. Performance is measured by **macro-averaged F1**, which gives equal weight to each class. Using accuracy would be misleading: a model always predicting the most common class (label 3, 93 images) scores 46.5% while being useless for others.

3.6. Cross-Validation Protocol

Both tasks use 5-fold Stratified K-Fold (`n_splits=5`, `shuffle=True`, `random_state=42`).

Per-fold pipeline:

1. Split 160 training / 40 validation, preserving class proportions.
2. Fit `StandardScaler` on training features; apply to both splits.
3. *Task B only:* apply SMOTE to the scaled training split.
4. Train model; evaluate on validation using the target metric.

We report mean $\pm$ standard deviation across five folds, alongside training scores as an overfitting reference. After CV-based model selection, the final model for each task (logistic regression for Task A; soft-voting ensemble for Task B) is retrained on all available training images using the same `StandardScaler` and SMOTE configuration fitted on the full training set, before generating held-out test predictions.

3.7. Bias–Variance Analysis

A large gap between training score and CV score signals overfitting (high variance); similar low values signal underfitting (high bias). Both Random Forest (CV 0.491) and XGBoost (CV 0.507) achieve perfect training macro-F1 (1.000), a clear overfitting signal; the train–CV gaps of $\Delta = 0.509$ and $\Delta = 0.493$ respectively motivate ensemble strategies.

4. Results

4.1. EDA: Feature Discriminability

Table 2 summarises Cohen's d for the key features from our 12-feature baseline EDA, which guided initial feature selection before we extended to the full 58-feature set.

LBP standard deviation at fine scale (`lbp_r1_std`) is the most discriminative single feature in both tasks. For Task A, Midwest Flooding images have higher LBP standard deviation (mean $= 3.046$) than SoCal Fire (mean $= 2.458$), consistent with greater fine-grained texture variability at the tile level in flood imagery. For Task B, destroyed images (label 3, mean $= 3.064$) show higher texture variability than minor-damage images (label 1, mean $= 2.693$), consistent with irregular rubble patterns; note these are tile-level associations that may partially reflect background land cover rather than isolated building damage.

FEATURE	CATEGORY	TASK A	TASK B
LBP_R1_STD	LBP	2.597	1.802
LBP_R3_MEAN	LBP	1.367	1.219
LBP_R2_MEAN	LBP	1.229	1.061
LBP_R2_STD	LBP	0.891	0.849
MEAN_B	COLOR	0.763	0.028
GABOR_STD	GABOR	0.455	0.008
EDGE_MEAN	EDGE	0.272	0.612

Table 2. Cohen's d for key features from the 12-feature baseline EDA (Task A: Flooding vs. Fire; Task B: Label 1 vs. 3). Values $d > 0.8$ indicate a large effect size.

Color features are useful for Task A — SoCal Fire images are consistently higher in mean blue intensity (mean $= 66.0$ vs. 54.3 for Midwest Flooding) — but nearly uninformative for Task B, where both labels come from the same disaster environment.

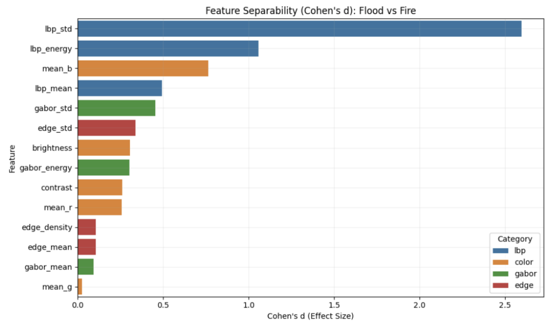
The extended EDA confirmed these patterns at multiple LBP radii: larger radii still separate classes, but `lbp_r1_std` remains dominant. For Gabor filters, response standard deviation outperforms response mean across all orientations and frequencies; lower spatial frequencies ($f = 0.1$–0.2) show stronger separation than $f = 0.3$ in both tasks.

The Pearson correlation matrix revealed strong intra-family redundancy: RGB means are nearly perfectly correlated with brightness; LBP mean and energy are correlated ($r \approx 0.85$); adjacent Gabor orientations share high inter-kernel correlation. This guided feature pruning in Section 3, yielding the 12-feature baseline. After observing suboptimal Task B performance with this baseline, we extended to the full 58-feature set (adding multi-radius LBP, full Gabor grid, and domain-motivated semantic features) as described in Section 2.4. Full Cohen's d rankings for all 58 features are in Appendix A. Note that Cohen's d was computed for the most practically relevant binary contrasts only (flooding vs. fire for Task A; label 1 vs. label 3 for Task B); pairwise d across all six Task B label pairs (e.g., label 0 vs. 2, label 1 vs. 2) would further reveal which class boundaries are hardest to separate.

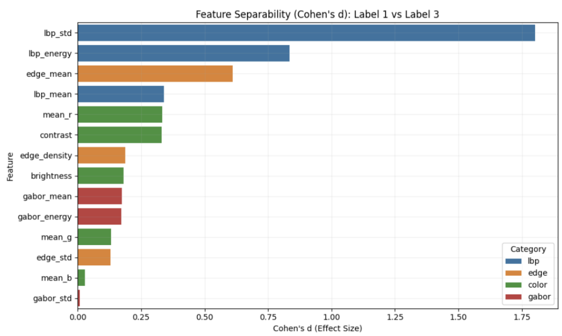
Figure 1 shows the ranked Cohen's d bar charts, and Figure 2 shows the best two-feature scatter plots for each task.

4.2. Task A: Classification Results

Logistic regression achieves near-perfect performance (Table 3), consistent with the large Cohen's d values for LBP features. The minimal training–CV accuracy gap indicates very low variance; the model generalizes well under default

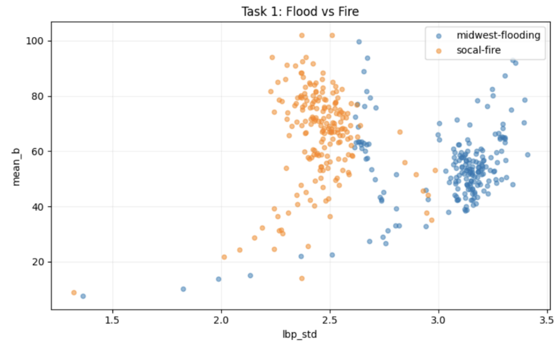


(a) Task A: Flooding vs. Fire

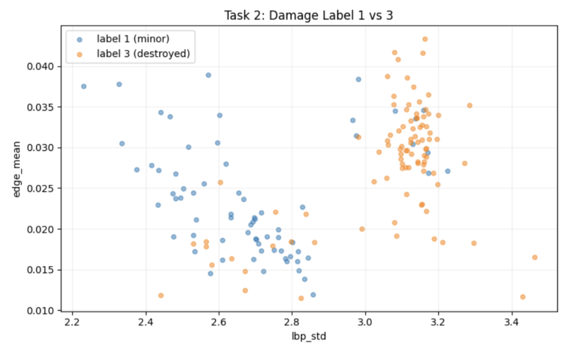


(b) Task B: Label 1 vs. 3

Figure 1. Feature discriminability (Cohen's d) for the 12-feature baseline. LBP features dominate both tasks. Color features rank higher for Task A than Task B.



(a) Task A: `lbp_std` vs. `mean_b`



(b) Task B: `lbp_std` vs. `edge_mean`

Figure 2. Best two-feature scatter plots. Midwest Flooding clusters at high `lbp_std`, low `mean_b`; SoCal Fire at the opposite. Label 3 clusters toward higher values on both axes.

L2 regularization ($C = 1.0$). On the held-out Pensieve test set the model scores **1.000 accuracy**, confirming that the LBP-driven linear boundary transfers perfectly to unseen flood and fire images.

METRIC	TRAIN	CV (5-FOLD)	TEST
ACCURACY	1.0000	0.9900 ± 0.0094	**1.0000**

Table 3. Task A logistic regression results. The minimal training–CV gap indicates little overfitting. Test accuracy reported by Pensieve.

4.3. Task B: Classification Results

Task B is significantly harder. Table 4 compares all models.

The best CV macro-F1 of 0.509 (soft ensemble, with GridSearch-tuned components) is well above random chance (0.25) but short of a fully operational system. Both Random Forest and XGBoost perfectly memorize the training set (Train F1 = 1.000), a clear overfitting signal; XGBoost's shallower trees (`max_depth=3`) yield better generalization, narrowing its train–CV gap to $\Delta = 0.493$ vs. RF's $\Delta = 0.509$ (Figure 3). Soft voting exploits the diversity among the three models, achieving the highest score (0.509) and lowest variance (± 0.044), indicating complementary failure modes. SMOTE unexpectedly reduced logistic regression performance: with only 6 label-2 training examples in Hurricane Matthew, interpolating between nearest neighbors generates synthetic samples in an extremely small region of feature space that likely overlaps with adjacent class distributions, shifting the decision boundary away from the true minority cluster. Tree-based models with class weighting are more robust to this phenomenon. Per-class precision, recall, and F1 across CV folds are reported in Appendix B; label 2 achieves zero F1 in all folds. Full hyperparameter settings are listed in Appendix D.

The soft ensemble was submitted to Pensieve and achieved a **weighted F1 of 0.677** on the held-out test set. This exceeds the CV macro-F1 of 0.509 because the weighted metric averages per-class F1 by support: with labels 1 (70 images) and 3 (93 images) well-represented and well-classified, their large support dominates the weighted average, whereas macro-F1 gives equal weight to label 2 (6 images) whose near-zero recall drags the macro score down significantly.

MODEL	TRAIN F1	CV MACRO-F1
LR, NO SMOTE	—	0.456 ± 0.053
LR, SMOTE (C=0.6)	0.626	0.449
RF (DEPTH=10, 100 TREES)	1.000	0.491
XGBOOST (LR=0.1, DEPTH=3)	1.000	0.507
SOFT ENSEMBLE	—	$\mathbf{0.509 \pm 0.044}$
Pensieve test set (soft ensemble): WEIGHTED F1 = **0.677**†		

Table 4. Task B model comparison (5-fold stratified CV macro-F1). SMOTE applied to training fold only. LR, RF, and XGBoost hyperparameters selected by `GridSearchCV`. RF train F1 of 1.000 vs. CV of 0.491 indicates memorization. † Pensieve reports weighted F1; it exceeds CV macro-F1 because majority classes (labels 1 and 3) dominate the weighted average while macro-F1 penalises near-zero label-2 recall equally.

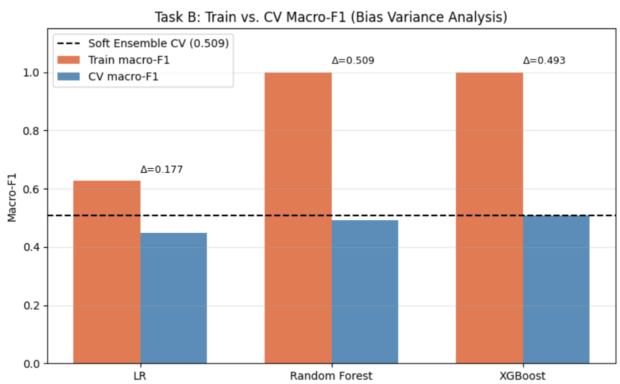


Figure 3. Train vs. CV macro-F1 for Task B models. Both RF and XGBoost perfectly memorize training data (Train F1 = 1.000); XGBoost's shallower trees (depth=3) achieve better CV generalization, giving a smaller gap ($\Delta = 0.493$) than RF ($\Delta = 0.509$). The dashed line marks the soft ensemble CV score (0.509).

5. Discussion

5.1. Findings and Interpretation

The central finding is a strong asymmetry between Task A and Task B. Disaster-type classification is effectively solved *under the current handcrafted feature representation and stratified CV protocol*: Midwest Flooding images contain large water bodies with high micro-texture variability, while SoCal Fire images show brighter, more uniform charred terrain. A linear boundary in the (`lbp_r1_std`, `mean_b`) feature space provides near-perfect separation, consistent with the large Cohen's d values in Table 2, and the Pensieve held-out test accuracy of 1.000 corroborates this within the current dataset. Generalisation to new geographic regions or disaster types would require evaluation beyond the current split design. Damage-severity classification is harder because the visual difference between minor damage and destruction within the same hurricane event is subtler, and the extreme scarcity of label 2 compounds the difficulty.

The dominance of LBP features over color and edge features is an important interpretive result. It suggests the discriminative signal lies not in the overall color of a tile but in the local microstructure of textures — the pattern of abrupt versus smooth pixel transitions at fine spatial scales. This aligns with physical intuition: destroyed structures produce irregular, heterogeneous rubble patterns, while intact surfaces tend to be spatially uniform.

5.2. Limitations

Data scarcity for Task B. With only 6 label-2 training examples, no oversampling strategy can compensate for the lack of real diversity. This is the fundamental bottleneck for damage severity classification.

Generalization to unseen data. Two distinct kinds of generalization should be distinguished. *Within-split generalization* (performance on held-out images from the same three events) is supported by our stratified CV results and confirmed by the Pensieve test scores. *Cross-disaster generalization*, covering entirely new disaster types such as earthquakes or tsunamis and different geographic regions, is not tested here. A disaster-level or region-level holdout would be needed to evaluate this more demanding form of transfer, and performance may degrade substantially on disaster types not present in training.

No spatial localization. All features are summary statistics computed over the full 1024×1024 tile. A tile containing one destroyed building surrounded by 50 intact structures will have global feature values nearly identical to a "no damage" tile. Detecting individual buildings and computing per-building features would dramatically improve sensitivity to localized damage, which the current global-statistics approach cannot address.

5.3. Societal Impacts and Ethical Considerations

Our system is designed to assist — not replace — human analysts in disaster response. **Asymmetric error costs.** Predicting minor damage when buildings are destroyed (false negative on label 3) may divert search-and-rescue teams away from the most critical areas. The reverse error wastes resources but does not cost lives. This asymmetric cost structure argues for conservative thresholds when deploying any model on label 3 predictions, and for reporting per-class recall alongside macro-F1.

Geographic and architectural bias. The xView2 training data covers specific US geographic regions and building typologies (wood-frame, brick, flat-roof construction common in the southeastern US and Iowa). A model trained on these data may generalize poorly to disasters in regions with different architectural styles, roof materials, or vegetation cover.

Failure mode on label 2. With only 6 training examples, any model is likely to have near-zero recall on major damage. Deployment requires a concrete triage strategy: (1) flag all images predicted as label 2; (2) flag any image where the two highest class probabilities differ by less than 0.15 (low-confidence predictions); (3) flag any image where the label 3 probability exceeds 0.3 even if the argmax is a lower label. All flagged images should receive mandatory human review before resource allocation decisions are made.

5.4. Future Work

Spatial localization. Our largest remaining limitation is the use of global tile-level statistics. Building-level feature extraction, which involves detecting individual structures and computing per-building descriptors, would dramatically improve sensitivity to localized damage and is the most impactful direction for future work.

Cost-sensitive learning. For disaster response, a false negative (predicting minor damage when buildings are destroyed) carries far higher real-world cost than a false positive. Optimising for recall of severe damage classes, rather than balanced macro-F1, may better reflect the operational context.

Lightweight CNN reference. A small CNN (e.g., fine-tuned MobileNetV2) could establish an upper bound on learned spatial features, making explicit the accuracy cost of our interpretable handcrafted approach.

Feature-family ablations. The final models use all 58 features, but the relative contribution of each family (LBP-only, color-only, Gabor-only, semantic/GLCM-only) is not isolated. Systematic ablations comparing single-family classifiers against the full-feature model would quantify how much each family independently contributes.

Building-count diagnostics. The `n_buildings` metadata column was excluded as a predictive feature, but it could serve as a diagnostic variable: stratifying error analysis by tile building count would reveal whether model failures concentrate in low-building or zero-building tiles, directly testing the hypothesis that global features fail when damage occupies only a small fraction of a tile.

6. Conclusion

We developed and evaluated an interpretable feature-based pipeline for post-disaster satellite image classification on the xView2 dataset. For disaster-type classification (Task A), logistic regression on the full 58-feature set achieves 99.0% cross-validated accuracy (98.75% on the 12-feature baseline), demonstrating that LBP texture features provide near-perfect separation between flood and fire imagery. For damage severity classification (Task B), a soft-voting ensemble of Logistic Regression, Random Forest, and XGBoost achieves a CV macro-F1 of 0.509 and a Pensieve test weighted F1 of 0.677, reflecting the combined challenge of limited feature discriminability and extreme class imbalance (label 2 has only 6 training examples).

Through our analysis, we can see that the choice of features matters as much as the choice of classifier. Systematic Cohen's d analysis and correlation-based pruning identified a minimal, non-redundant feature set that captures most of the cross-validated performance. Going forward, empirical feature importance re-selection, richer spatial representations, and alternative oversampling strategies are our immediate priorities before exploring more complex ensemble models.

A video presentation summarising this work is available at: https://youtu.be/tSi-gjOoNSM.

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7. Appendix

A. Full Feature Discriminability Rankings

Table 5 and Table 6 report Cohen's d for all 58 features, ranked separately by Task A and Task B discriminability. These values motivated the transition from the 12-feature baseline to the full 58-feature set. Semantic features (gray_pixel_ratio, laplacian_var, veg_index_std) rank among the top features for both tasks, confirming their value beyond generic image statistics.

Feature	Category	Task A	Task B
LBP_R1_STD	LBP	2.597	1.802
GRAY_PIXEL_RATIO	Semantic	2.529	0.220
SAT_MEAN	Color	1.543	0.190
LBP_R3_MEAN	LBP	1.367	1.219
VEG_INDEX_STD	Semantic	1.313	0.548
LBP_R2_MEAN	LBP	1.229	1.061
LBP_R2_STD	LBP	0.891	0.849
GLCM_CORRELATION	Semantic	0.829	0.181
LAPLACIAN_VAR	Semantic	0.822	1.489
MEAN_B	Color	0.763	0.028
IQR_G	Color	0.647	0.137
IQR_B	Color	0.645	0.045
GABOR_T135_F02_STD	Gabor	0.637	0.218
GABOR_T135_F01_STD	Gabor	0.632	0.219
GABOR_T90_F01_STD	Gabor	0.630	0.218
GABOR_T0_F01_STD	Gabor	0.630	0.218
STD_G	Color	0.629	0.115
GABOR_T45_F01_STD	Gabor	0.628	0.217
VEG_INDEX_MEAN	Semantic	0.535	0.360
EDGE_MEAN	Edge	0.105	0.612

Table 5. Top-20 features ranked by Task A Cohen's d (Flooding vs. Fire). Task B values shown for comparison. Gabor features cluster tightly around $d \approx 0.63$ for Task A but contribute little to Task B.

Feature	Category	Task A	Task B
LBP_R1_STD	LBP	2.597	1.802
LAPLACIAN_VAR	Semantic	0.822	1.489
LBP_R3_MEAN	LBP	1.367	1.219
LBP_R2_MEAN	LBP	1.229	1.061
LBP_R2_STD	LBP	0.891	0.849
EDGE_MEAN	Edge	0.105	0.612
VEG_INDEX_STD	Semantic	1.313	0.548
HUE_MEAN	Color	0.024	0.504
SAT_STD	Color	0.372	0.463
BLUE_TARP_RATIO	Semantic	0.048	0.444
STD_R	Color	0.287	0.419
HUE_STD	Color	0.165	0.392
VEG_INDEX_MEAN	Semantic	0.535	0.360
LBP_R1_MEAN	LBP	0.494	0.339
MEAN_R	Color	0.256	0.332
GLCM_CONTRAST	Semantic	0.121	0.324
GLCM_HOMOGENEITY	Semantic	0.110	0.243
GRAY_PIXEL_RATIO	Semantic	2.529	0.220
GABOR_T135_F01_STD	Gabor	0.632	0.219
IMAGE_ENTROPY	Semantic	0.313	0.213

Table 6. Top-20 features ranked by Task B Cohen's d (Label 1 vs. 3 in Hurricane Matthew). Task A values shown for comparison. blue_tarp_ratio ($d = 0.444$) and hue_mean ($d = 0.504$) are task-specific signals with near-zero discriminability for Task A.

B. Task B Per-Class Performance

Table 7 reports precision, recall, and F1 per damage level for the Random Forest model, aggregated across all five CV folds. The results reveal the underlying difficulty obscured by the macro-F1 summary.

The zero F1 on label 2 is the central failure mode of our pipeline. It reflects a fundamental data limitation: with only 6 training examples, each fold leaves at most 4–5 label-2 examples for training, making it impossible to learn a reliable decision boundary. The model defaults to never predicting major damage, instead assigning label-2-like tiles to the adjacent labels 1 or 3. This is precisely the failure mode that motivates collecting additional label-2 training data before any further modeling effort.

Label	Precision	Recall	F1	Support
0 — No damage	0.44	0.52	0.48	31
1 — Minor	0.65	0.61	0.63	70
2 — Major	0.00	0.00	0.00	6
3 — Destroyed	0.83	0.85	0.84	93
Macro avg	0.48	0.49	0.49	200
Weighted avg	0.68	0.69	0.69	200

Table 7. Random Forest per-class performance (aggregated over 5 CV folds). Label 2 achieves zero F1; the model never correctly predicts "major damage" on held-out data. Label 3 achieves 0.84 F1 due to its large support; label 1 achieves 0.63.

C. Random Forest Feature Importances

Table 8 reports mean impurity decrease (Gini importance) for the top-20 features in the Random Forest trained on all 200 Hurricane Matthew images (full 58-feature set, SMOTE applied). Feature importances are complementary to Cohen's d: while d measures univariate separability, importances reflect each feature's marginal contribution within the full ensemble of trees.

D. Model Hyperparameter Reference

Table 9 lists all hyperparameters used in the final experiments for reproducibility.

FEATURE	CATEGORY	IMPORTANCE
VEG_INDEX_STD	SEMANTIC	0.0715
LAPLACIAN_VAR	SEMANTIC	0.0480
LBP_R1_STD	LBP	0.0452
LBP_R3_MEAN	LBP	0.0412
LBP_R2_MEAN	LBP	0.0380
GLCM_CORRELATION	SEMANTIC	0.0366
BLUE_TARP_RATIO	SEMANTIC	0.0334
LBP_R2_STD	LBP	0.0330
SAT_MEAN	COLOR	0.0291
VEG_INDEX_MEAN	SEMANTIC	0.0289
LBP_R1_MEAN	LBP	0.0284
HUE_MEAN	COLOR	0.0281
EDGE_MEAN	EDGE	0.0275
HUE_STD	COLOR	0.0263
GRAY_PIXEL_RATIO	SEMANTIC	0.0225
LBP_R3_STD	LBP	0.0208
GLCM_HOMOGENEITY	SEMANTIC	0.0201
SAT_STD	COLOR	0.0198
GLCM_ENERGY	SEMANTIC	0.0184
EDGE_STD	EDGE	0.0183

Table 8. Random Forest feature importances (mean impurity decrease) for Task B, top 20 of 58 features. `veg_index_std` ranks first by importance despite ranking 7th by Cohen's d, reflecting its marginal contribution in multi-feature context. Semantic features occupy 4 of the top 6 positions, validating the domain-motivated feature engineering effort.

MODEL	HYPERPARAMETER	VALUE
LOGISTIC REGRESSION (TASK B)	C	0.6 (ITERATIVE TUNING)
	MAX_ITER	1000
	CLASS_WEIGHT	BALANCED
RANDOM FOREST	N_ESTIMATORS	100 (GRIDSEARCH BEST)
	MAX_DEPTH	10 (GRIDSEARCH BEST)
	CLASS_WEIGHT	BALANCED
	RANDOM_STATE	42
XGBOOST	N_ESTIMATORS	300 (GRIDSEARCH BEST)
	MAX_DEPTH	3 (GRIDSEARCH BEST)
	LEARNING_RATE	0.1 (GRIDSEARCH BEST)
	RANDOM_STATE	42
SMOTE	K_NEIGHBORS	3
	RANDOM_STATE	42
CROSS-VALIDATION	N_SPLITS	5
	SHUFFLE	TRUE
	RANDOM_STATE	42

Table 9. Hyperparameter reference for all models. `class_weight=balanced` sets $w_c = n/(n_{\text{classes}} \cdot n_c)$. All random states fixed to 42 for reproducibility.